

MASTER PROJECT

Shock propagation through the lens of remittances

Who bears the cost? Evidence from US–Mexico migration corridors

Authors

Anqi Wang Liu, Francesco Zucca, Joan Costa Rodríguez,

Michael Fehl, Xènia Alemany Rovira

Barcelona School of Economics

Master's program in Economics

Abstract

This paper studies how localized shocks in migrant destination areas propagate to origin communities through remittance networks. We use Hurricane Ian, which struck Florida in 2022, as a spatially concentrated negative shock to remittance senders in the United States. Combining municipality-level remittance data from Banco de México with migration-network measures based on Mexican consular records, we examine whether Mexican municipalities more exposed to Florida experienced differential changes in remittance inflows after the shock. We then use the structure of migrant networks to study whether exposure to other major U.S. destinations, especially Texas and California, mitigated or amplified the shock. The results speak to how international remittance networks transmit local economic disruptions across borders and to the role of migrant-network diversification in shaping the distribution of these effects.

Keywords: Remittances, migration networks, localized shocks, Hurricane Ian, Mexico, spillover effects

Resum

Aquest treball investiga com els xocs localitzats a les àrees de destinació dels migrants es propaguen cap a les comunitats d'origen a través de les xarxes de remeses. Utilitzem l'huracà Ian, que va colpejar Florida l'any 2022, com un xoc negatiu espacialment concentrat sobre els emissors de remeses als Estats Units. Combinant dades de remeses a escala municipal del Banco de México amb mesures de xarxes migratòries basades en registres consulars mexicans, examinem si els municipis mexicans amb una exposició més gran a Florida van experimentar canvis diferencials en les entrades de remeses després del xoc. A continuació, utilitzem l'estructura de les xarxes migratòries per estudiar si l'exposició a altres grans destinacions dels Estats Units, especialment Texas i Califòrnia, va mitigar o amplificar el xoc. Els resultats aporten evidència sobre com les xarxes internacionals de remeses transmeten perturbacions econòmiques locals a través de les fronteres i sobre el paper que té la diversificació de les xarxes migratòries en la distribu-

ció d'aquests efectes.

Paraules clau: Remeses, xarxes migratòries, xocs localitzats, huracà Ian, Mèxic, efectes de propagació

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1. Introduction

International remittances are one of the main private channels through which local economic shocks can cross national borders. Mexico is a central setting in which to study this mechanism. Banco de México records US\$64.7 billion in family remittance receipts in 2024, while World Bank estimates place Mexico among the largest remittance recipients in the world, second only to India (Banco de México, 2026b; World Bank, 2024). The U.S.–Mexico corridor is especially important because Mexican households receive transfers from migrants whose earnings are tied to local labor markets across the United States. This paper asks whether a negative local shock in the United States is transmitted to Mexican municipalities through remittances. Do migrants smooth transfers after being hit by a shock, or do origin households bear part of the adjustment through lower remittance inflows?

The answer matters because remittances are not only an aggregate balance-of-payments flow, they are also part of household resources. Our data do not directly observe Mexican household income or consumption. Even so, changes in remittance inflows are economically meaningful for household welfare. Prior work shows that remittances are an important source of household resources in Mexico, affecting consumption, poverty, and inequality (Taylor et al., 2005), food security (Mora-Rivera & van Gameren, 2021), and schooling and child labor decisions (Alcaraz et al., 2012). We therefore interpret changes in municipal remittance inflows as reduced-form evidence on a channel with direct relevance for household income and consumption, while not claiming to estimate those household outcomes themselves.

The main empirical difficulty is that most large shocks are not cleanly assigned to one side of the migration corridor. Caballero et al. (2023) show that Mexican households connected to U.S. labor markets more severely affected by the Great Recession were less likely to receive remittances, providing important evidence that migrant networks transmit U.S. shocks to Mexico. Yet the Great Recession simultaneously affected the United States and Mexico, making it difficult to isolate the remittance channel from other aggregate forces. Their analysis also focuses on the extensive margin of remittance receipt. Quantifying how much bilateral remittance flows adjust on the intensive margin is nec-

essary to measure the income loss experienced by origin households.

We address these challenges by exploiting Hurricane Ian, which made landfall in southwestern Florida on September 28, 2022. Ian caused an estimated US\$109.5 billion in damages in Florida and was the costliest hurricane in the state's history (Bucci et al., 2026). The shock was severe and spatially concentrated on the sender side of the corridor: it disrupted economic activity in Florida but did not directly affect Mexican municipalities. Natural disasters in the United States can reduce local productivity, labor demand, housing wealth, and household resources (Boustan et al., 2020). Hurricane Ian therefore provides a localized negative shock to the capacity of Florida-based migrants to remit, while leaving recipient municipalities in Mexico exposed only through their pre-existing migration links to Florida.

We base our argument on the following idea. As a response to a shock to Florida, migrants can directly reduce remittances from Florida to Mexican municipalities. However, many origin municipalities receive remittances from migrants in several U.S. states. If migrants care about total resources available to the origin household, then a fall in Florida remittances can raise the marginal value of transfers from migrants living elsewhere. The same shock can therefore generate two effects at once: a direct decline in Florida-to-Mexico remittances and a compensatory response from other U.S. states connected to the same municipalities. An alternative behaviour of the migrants in the US could also be to domestically remit to the affected state households instead of counteracting the shock through remittances to Mexico. This response assumes that other state migrants are connected to the Florida mexicans. This two reactions may be mutually exclusive, pushing the change in remittances recieved by Mexican municipalities onto opposite directions. The net effect on remittance inflows to Mexican municipalities is therefore an empirical question this paper addresses.

The first part of the paper formalizes this idea with a structural remittance model. The model follows the insight in Albert and Monras (2022) that migrants allocate resources across where they live and where their origin households are located. In our setting, this idea is applied to remittance choices rather than to location choices. Migrants choose between local consumption in the United States and resources available to the origin

household in Mexico. They differ in income and in the strength of their altruistic or obligation motive, and remitting involves both a fixed cost and a variable cost. In the spirit of models with heterogeneous agents, fixed costs, and variable costs (Melitz, 2003), the fixed cost generates an extensive margin of remitting, while the variable cost governs the intensity of transfers.

The key extension is that origin-household resources include local non-remittance income plus remittances received from migrants in other U.S. states. This makes the model directly useful for interpreting the empirical patterns. A negative shock to Florida migrants reduces their optimal remittances. At the same time, for households linked to Florida, the fall in Florida transfers lowers perceived origin resources for migrants in other states, raising the marginal utility of additional remittances under an insurance motive. The model therefore provides a unified way to interpret both the direct exposure effect and the compensatory network response observed in the data.

The second part of the paper brings the model to the data by constructing the inflow of remittances flows from U.S. states to Mexican municipalities. Because such flows are not directly observed, we combine two sources from Banco de México with MCAS administrative records. Banco de México reports remittance inflows by U.S. state of origin and by Mexican municipality of receipt (Banco de México, 2026a, 2026c). MCAS records identify the Mexican municipality of origin and U.S. state of residence of Mexican consular-card holders, providing a proxy for the geography of migrant networks. We use these records to construct state-to-municipality migration shares and allocate observed state-level remittance outflows across Mexican municipalities. We focus our analysis on the Mexican municipalities receiving remittances from Florida.

Two validation exercises support this construction. First, the migration network data are highly persistent: the distribution of Mexican migrants across U.S. states and Mexican municipalities has year-to-year correlations above 0.90 from 2012 onward. Second, when we use these migration shares to allocate state-level remittance outflows to Mexican municipalities, the resulting municipal aggregates correlate between 0.75 and 0.85 with official municipal remittance receipts. These facts suggest that the estimated corridors capture meaningful variation in the geography of remittances and that the assumptions

underlying our construction are reasonable.

The third part of the paper studies the shock. We exploit variation in Mexican municipalities' pre-existing exposure to Florida to estimate whether Ian affected municipal remittance inflows differently depending on their exposure to the shocked state. In the period immediately following Ian, a 1% increase in a municipality's pre-existing exposure to Florida is associated with a 0.16% decrease in remittances received. This result is consistent with a sender-side income shock being transmitted through the remittance corridor to origin municipalities.

Finally, we examine migration-network spillovers. Among municipalities jointly exposed to Florida and other major sender states, remittances originating from California remain largely unchanged following the shock. By contrast, remittances from Texas, the second-largest sender state, decline in municipalities that are jointly exposed to both migration corridors in the post-Ian period. We explore this relationship further by centering the data on different quantiles of Texas exposure to Mexican municipalities. The diagnostic plots show for the 10% most exposed Florida municipalities to Texas, (insert write-up here), whereas for the least 10% exposed show (describe what we see). This pattern is consistent with both compensatory transfers from migrants in other states and out-migration from Florida-exposed counties.

Related Literature. The paper is close to work that studies remittances as a form of household insurance (Amuedo-Dorantes & Pozo, 2006; Bettin et al., 2025; Yang & Choi, 2007). Most existing evidence studies shocks in receiving countries, where migrants abroad may increase transfers to insure households at origin. We instead study a sender-side shock, where the capacity to remit changes in the destination labor market while origin municipalities are not directly affected. This distinction is important because the same remittance channel that insures households against origin shocks may transmit destination shocks back to those households.

The paper also relates to research on migrant networks and the international transmission of local shocks (Caballero et al., 2023; Munshi, 2003). The closest paper is Caballero et al. (2023), which shows that Mexican households linked to U.S. labor markets hit by

the Great Recession became less likely to receive remittances. We differ by studying a localized natural disaster, constructing bilateral state-to-municipality remittance corridors, and focusing on intensive-margin flows and cross-corridor responses.

The structural framework connects to research that models migrants as allocating resources across origin and destination locations (Albert & Monras, 2022). The difference is that our model uses this logic to study remittance behavior after a destination-side shock. It also connects to work on interference and spillovers in causal inference (Aronow & Samii, 2017; Butts, 2023; Hudgens & Halloran, 2008). In our setting, interference is not only a threat to identification, but part of the economic mechanism, because migrants linked to the same origin municipality may jointly insure households against shocks that affect only one destination.

The remainder of the paper is organized as follows. Chapter 2 develops the theoretical remittance model. Chapter 3 introduces Hurricane Ian as the natural experiment and describes both the construction and validation of the migration and remittance data. Chapter 4 presents the empirical strategy and reports the main estimates for exposure effects. Chapter 5 investigates potential spillover effects. Chapter 6 concludes.

2. Theoretical framework and empirical strategy

This chapter outlines the theoretical framework used for our analysis. We begin by adapting a heterogeneous-firm trade model to the context of individual remittance behavior, establishing the theoretical foundations of how a migrant's transfer decisions can be influenced by economic shocks. Then, we expand the model by allowing for spillover effects across migrant networks. Finally, we provide the main predictions of the model, which guide our empirical strategy in the subsequent chapters.

2.1. A Remittance Model with Heterogeneous Migrants

To analyze the impact of shocks on remittance flows, we adapt the logic of heterogeneous-firm trade models to the context of remittances. Specifically, we consider a model where migrants differ in income and altruism intensity, and where remitting requires paying both a fixed access cost and a variable cost proportional to the amount remitted. For-

mally, consider migrant i , who resides in U.S. state s , originates from Mexican municipality m , and chooses remittances in period t . The migrant's utility function is given by:

$$U_{imst} = \left[\alpha \left(c_{imst}^L \right)^{\frac{\sigma-1}{\sigma}} + (1 - \alpha) \left(\phi_{im} [R_{imst} + y_{mt}^O] \right)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \quad (2.1)$$

Here $\phi_{im} > 0$ denotes the altruism or obligation intensity parameter, analogous to the productivity parameter in Melitz (2003). This strictly positive parameter ensures that sending remittances can increase the migrant's utility by raising origin-household resources. The variable $c_{imst}^L \geq 0$ represents the migrant's consumption in the destination country, and $R_{imst} \geq 0$ is the amount sent in remittances by migrant i . The weight of local consumption is given by $\alpha \in (0, 1)$, and the elasticity of substitution between local consumption and origin-household resources is given by σ . Origin-household total resources are therefore defined as $R_{imst} + y_{imst}^O$, where the presence of y_{imst}^O introduces a role for origin-country economic conditions: lower perceived origin resources increase the marginal utility of remittances, generating an insurance motive consistent with Yang and Choi (2007).

The migrant's budget constraint is given by:

$$c_{imst}^L + f_{mt} \mathbb{1}\{R_{imst} > 0\} + (1 + \tau_{mt})R_{imst} \leq w_{imst} \quad (2.2)$$

where $w_{imst} \geq 0$ is the migrant's wage, $f_{mt} \geq 0$ is a fixed remitting cost capturing the pre-transaction expense of accessing a remittance channel, analogous to the fixed export cost in Melitz (2003), and $\tau_{mt} > -1$ is the variable cost which scales with the amount remitted, analogous to the iceberg cost structure in Melitz (2003). Notice that, even though τ_{mt} is typically positive to reflect transaction fees or exchange rate spreads, we allow for the possibility of negative variable costs, which could arise from the presence of remittance subsidies or favorable exchange rate regimes.

If the migrant maximizes utility subject to the budget constraint and chooses to remit a positive amount ($R_{imst} > 0$), solving the Lagrangian yields the optimal remittance

function:

$$R_{imst}^* = \frac{(w_{imst} - f_{mt}) - \frac{\phi_{im}}{A_{imst}} y_{mt}^O}{\frac{\phi_{im}}{A_{imst}} + (1 + \tau_{mt})} \quad (2.3)$$

where A_{imst} represents the optimal ratio of origin-household resources to own local consumption, and is defined as:

$$A_{imst} \equiv \left(\frac{(1 - \alpha)\phi_{im}}{\alpha(1 + \tau_{mt})} \right)^\sigma \quad (2.4)$$

To understand the effect of destination-country income shocks on remittance flows, we take the partial derivative of optimal remittances with respect to wage:

$$\frac{\partial R_{imst}^*}{\partial w_{imst}} = \frac{A_{imst}}{1 + (1 + \tau_{mt})A_{imst}} > 0 \quad (2.5)$$

Since the derivative is strictly positive, through an income effect the model predicts that a positive shock to the migrant's wage will increase remittances, while a negative shock will decrease them. The fixed cost f_{mt} establishes an entry threshold: if the migrant's wage is not sufficiently high relative to the fixed cost, they will not send remittances at all. Additionally, the variable cost τ_{mt} reduces the marginal benefit of each dollar sent, dampening the sensitivity of remittances to wage fluctuations.

2.2. Allowing for Spillovers

We expand the model to allow for spillover effects across migrant networks. Decomposing origin-household resources into local non-remittance income and remittances received from migrants in other US states, the perceived origin-household resources excluding migrant i 's own remittances can be expressed as:

$$y_{imst}^O = \tilde{y}_{mt}^O + \sum_{s' \neq s} R_{s'mt} \quad (2.6)$$

Solving for optimal remittances in the augmented model yields:

$$R_{imst}^* = \frac{(w_{imst} - f_{mt}) - \frac{\phi_{im}}{A_{imst}} (\tilde{y}_{mt}^O + \sum_{s' \neq s} R_{s'mt})}{\frac{\phi_{im}}{A_{imst}} + (1 + \tau_{mt})} \quad (2.7)$$

Hence, holding the migrant's own wage and remitting costs fixed, remittances from other migrants enter the decision through perceived origin-household resources:

$$\frac{\partial R_{imst}^*}{\partial (\sum_{s' \neq s} R_{s'mt})} = - \frac{\phi_{im}}{\phi_{im} + A_{imst}(1 + \tau_{mt})} \quad (2.8)$$

Because the derivative is strictly negative, high exposure to a destination shock lowers perceived origin resources and raises the marginal utility of alternative transfers. This drives compensatory network spillovers, where the direct remittance drop generates an offsetting increase from other destinations.

To ground these concepts, the model delivers two empirical implications that guide our analysis. First, because optimal remittances are increasing in migrant income, a negative wage or income shock in the destination country should reduce remittances sent by affected migrants. Second, because remittances from other destinations enter the migrant's decision through perceived origin-household resources, a decline in transfers from the affected destination raises the marginal value of remittances from migrants in alternative destinations connected to the same municipality. Hence, origin municipalities with greater pre-existing exposure to the affected migrant destination should experience larger declines in total remittance receipts, all else equal. Based on these predictions, we develop our empirical strategy as follows: first, we test whether local inflows fall more in Mexican municipalities highly exposed to the destination hit by the shock; then, we examine whether co-exposure to other high-sending US states not directly affected mitigate the negative impact on total remittance revenue.

3. Empirical Context and Data

This chapter introduces the empirical context and data used in our analysis. We begin by introducing our source of exogenous variation in remittance flows, Hurricane Ian. First,

we document the hurricane’s impact on Florida’s remittance outflows as well as sectors with high Mexican immigrant employment. Next, we explain the data sources and the methodology we use to construct our treatment measure, which differentiates Mexican municipalities by how exposed they are to the shock. Finally, we present a validation exercise, justifying the choice of our constructed treatment variable as a proxy for the Mexican migrant population in the United States.

3.1. The Natural Experiment: Hurricane Ian (2022)

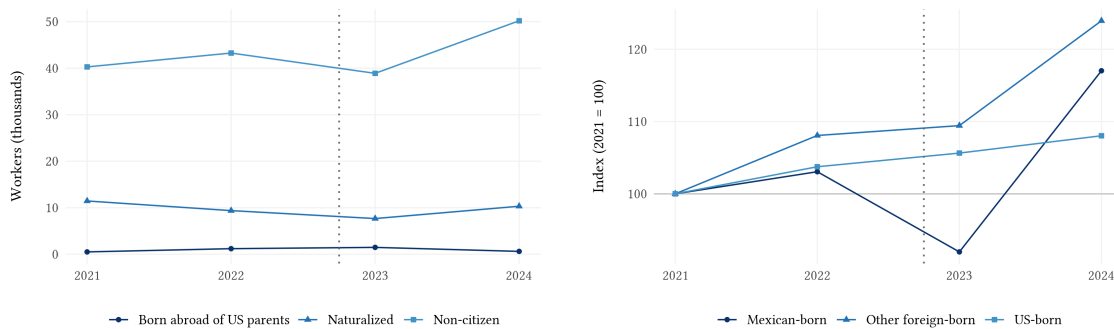
Hurricane Ian struck the southwestern coast of Florida on September 28, 2022, with peak sustained winds of 140 kt (around 260 kph) (Bucci et al., 2026). Ian caused an estimated US\$ 112.9 billion in total damage in the United States, of which US\$ 109.5 billion occurred in Florida alone, making it simultaneously the costliest hurricane in Florida’s history and the third-costliest in US history. In Lee County alone, at least 52,514 structures were impacted, of which 5,369 were completely destroyed (Bucci et al., 2026).

Beyond its aggregate toll, Ian’s destruction fell on sectors that employ a large share of Mexican migrants. Construction is the single largest sector of employment for Mexican-born workers in Florida, at roughly 28% of all those employed, and the four sectors most exposed to the storm together account for over 50% of Mexican workers. Within them, construction makes up 52%, agriculture 17%, landscaping 16%, and hospitality 15%.¹ Construction is therefore both the main source of work for Mexican workers and the channel where Ian’s damage translates most directly into lost employment, and later into reconstruction demand.

Figure 3.1b indexes Florida construction employment to 2021 by workers’ region of birth. The Mexican-born series diverges from the US-born and other foreign-born series after Ian, falling to roughly 92% of its 2021 level in 2023 before rebounding to about 117% in 2024. The US-born series drifts upward throughout, which suggests the disruption was specific to migrant labor rather than to construction as a whole. Figure 3.1a shows that this movement is carried by non-citizens, who make up the large majority of Mexican-

¹Calculated using IPUMS USA, ACS 1-year samples 2021–2024, employed Mexican-born residents of Florida. The full industry composition and the within-sector shares are reported in the online appendix.

born construction workers in the state and absorb both the 2023 dip and the 2024 rebound. Non-citizens are also the group most reliant on remittances as a financial link to their origin households (Handlin et al., 2002). The descriptive evidence thus indicates that Ian affected workers whose earnings finance remittances to Mexico, which motivates the migrant income channel we examine in the remainder of the paper.



(a) Mexican-born construction workers by citizenship status

(b) Construction employment by region of birth, indexed to 2021

Figure 3.1: Mexican-born construction employment in Florida around Hurricane Ian. Source: IPUMS USA, ACS 1-year samples 2021–2024; employed Florida construction workers. Dotted line marks Hurricane Ian (2022Q4).

Crucially for our identification strategy, Hurricane Ian did not affect Mexico. The storm originated in the Caribbean, made landfall in Cuba and Florida, and dissipated over the Atlantic after a final less impactful landfall in South Carolina (Bucci et al., 2026). Therefore, the shock is spatially clean, allowing us to better isolate the transmission mechanism through the remittance channel. In the language of the model proposed in Chapter 2, Hurricane Ian is the negative shock to migrants’ economic resources, Florida is the affected destination, and Mexican municipalities differ in their pre-existing exposure to Florida through migration networks.

To motivate the analysis, Figure 3.2 presents the evolution of remittance outflows from Florida to Mexico using data from Banxico series CE168 (Banco de México, 2026a). A sharp decline in remittance outflows can be observed in the quarter immediately following Hurricane Ian, with flows falling by approximately 40% relative to the previous quarter. This pronounced contraction suggests that the hurricane generated a substantial disruption in remittance activity from Florida to Mexico, thereby providing a strong

motivation for examining its effects on Mexican municipalities' remittance inflows.²

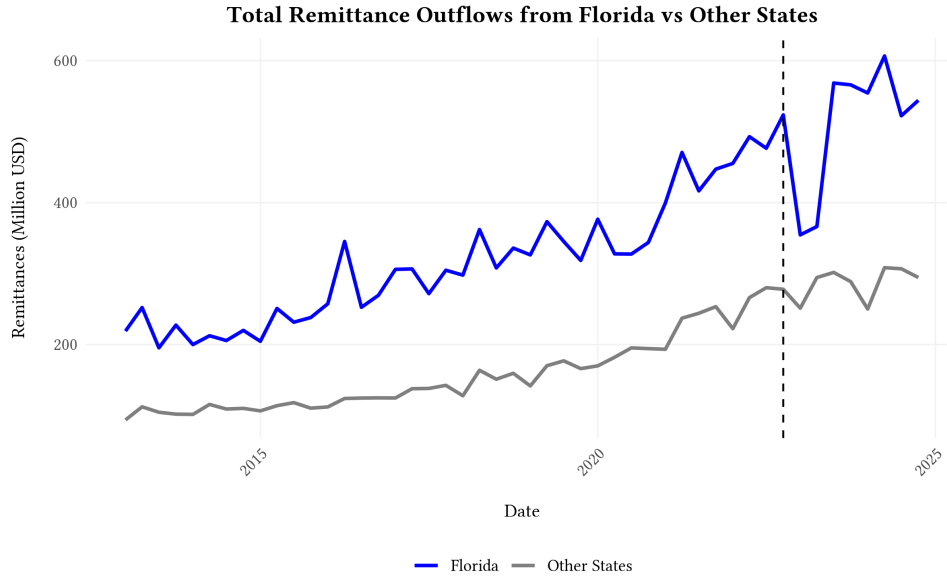


Figure 3.2: Remittance outflows from Florida to Mexico, in millions of US dollars. The vertical dashed line indicates the quarter of Hurricane Ian (Q4 2022).

Source: Authors' own elaboration based on Banxico SIE data.

3.2. Data Sources

Given that bilateral remittance flows between US States and Mexican municipalities are not publicly available, we base our estimation strategy on multiple data sources. The source of remittance data is Banco de México (Banxico), through the datasets provided by its Sistema de Información Económica (SIE). Specifically, series CE166 records remittance inflows received by each Mexican municipality. This series is available at quarterly frequency, from Q1 2013 to Q1 2026 and represents our outcome variable (Banco de México, 2026c). To capture the Mexican migration network, we utilize administrative records from the Matricula Consular de Alta Seguridad (MCAS) program, managed by the Instituto de los Mexicanos en el Exterior (IME) (Instituto de los Mexicanos en el Exterior, 2026). The publicly available version of this dataset provides a comprehensive registry of consular identification cards issued to Mexican nationals residing in the United States

²Florida outflows are plotted against the average outflows of all other U.S. states. Because the comparison series is constructed as an average, potential opposite shocks affecting individual states during the Hurricane Ian period may be attenuated. Nevertheless, the purpose of the figure is simply to illustrate the magnitude of the decline observed in Florida.

each year, from 2010 to 2024. While the MCAS does not constitute an exhaustive census of all migrants, it represents a robust proxy for the spatial distribution of Mexican migrants in the United States. In particular, it is highly representative of undocumented and recently arrived migrants, who face the strongest incentives to obtain consular identification (Caballero et al., 2018).

3.3. Migration Networks Construction

To build our treatment measure, we construct a migration-based matrix using annual MCAS data from 2014 to 2021. Because these records capture new consular issuances rather than the complete stock of the remitting population, which also includes previously arrived migrants, we rely on the assumption of persistence of migration patterns over time (Card, 2001; Munshi, 2003). Additionally, we rely on the fact that there is no significant internal mobility within Mexico, allowing us to link remittance receiving municipalities to the migrants' origin villages. This is supported by the literature, showing that internal mobility has decreased over the last few decades (Gordillo de Anda & Plassot, 2017; Partida Bush, 2014).

To mitigate the volatility of yearly migration fluctuation and construct a robust proxy for the long-term spatial distribution of Mexican migrants, we compute the average of the migration matrices across all years. Formally, we define $MCAS_{ms}$ as the average annual count of MCAS issuances to migrants from Mexican municipality m residing in U.S. state s . For each Mexican municipality, we calculate the average migration network share, w_{ms} , which represents the proportion of municipality m 's total US migrant population that resides in state s . This is calculated as the ratio of $MCAS_{ms}$ to the total number of MCAS issuances for migrants from municipality m across all U.S. states S :

$$w_{ms} = \frac{MCAS_{ms}}{\sum_{s=1}^S MCAS_{ms}} \quad (3.1)$$

By construction, the sum of w_{ms} across all U.S. states for a given Mexican municipality m equals 1, and the value of w_{ms} for a specific state s reflects the relative importance of that state as a destination for migrants from municipality m , which in turn is a proxy for the exposure of municipality m to shocks occurring in state s .

3.4. Persistence Assumption and Validation

The validity of this estimation relies on the assumption of temporal persistence of migration flows, implying that the distribution of migrants recorded in the MCAS data is representative of the active remitting population. The stability of migration patterns over time is well-documented in the literature, with studies showing that existing migration networks tend to dictate future migration patterns, creating persistent flows between specific origin and destination pairs (Card, 2001; Munshi, 2003). To verify this assumption within our specific sample period, we conduct a validation exercise using MCAS data. Figure 3.3 computes the Pearson correlation of our constructed weighting matrices across years. As observed, when taking any year from 2014 onward as the base year, the correlation with subsequent years is consistently above 0.90.

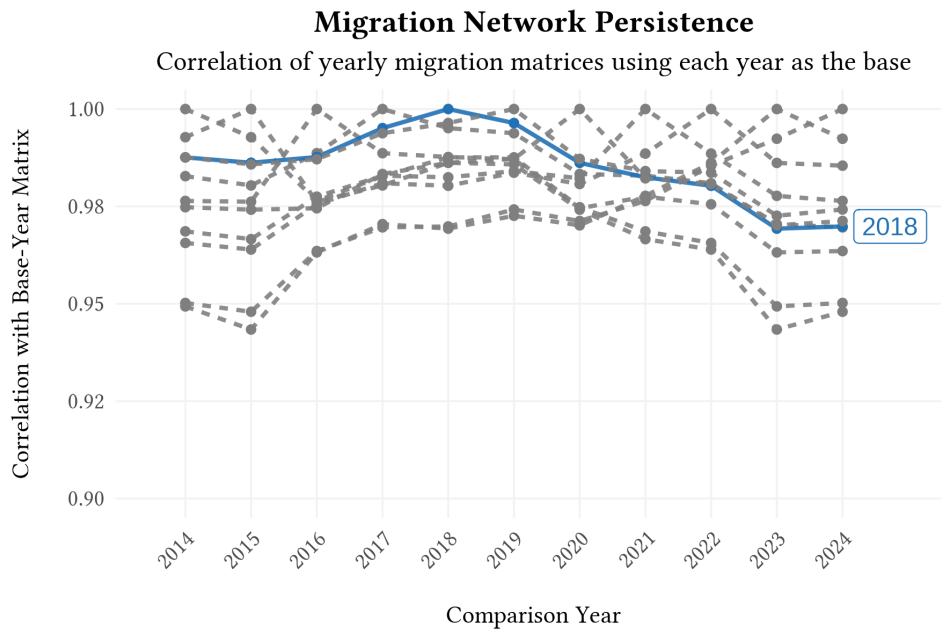


Figure 3.3: Correlation of MCAS-based weighting matrices across years. Each point represents the correlation coefficient between the weighting matrix of the base year and the weighting matrix of the comparison year (2018 as base year is highlighted in blue for demonstration purposes).

Source: Authors' own elaboration based on MCAS data.

This high degree of correlation suggests that the spatial distribution of Mexican migrants across U.S. states has remained highly stable over time, lending credibility to our use of an averaged weighting matrix for estimating bilateral remittance flows. Given that Hurricane Ian hit Florida in Q4 2022 and the MCAS data for Florida in 2013 is missing,

to construct our treatment variable we use the average of the weighting matrices from 2014 to 2021, which are all highly correlated with each other, and avoid any endogeneity in the treatment caused by the shock.

4. Empirical Methodology and Results

4.1. Estimation Strategy

The primary challenge in estimating the impact of the Hurricane Ian shock on remittance inflows to Mexican municipalities lies in the sorting of migrants. Municipalities with higher exposure to Florida may differ systematically from those with networks concentrated in other U.S. states. Therefore, a standard cross-sectional analysis comparing municipalities with different levels of exposure to Florida would likely lead to biased estimates due to confounding factors correlated with both the treatment (exposure to Florida) and the outcome (remittance inflows). This classic omitted variable bias problem makes a causal interpretation of cross-sectional estimates impossible (Wooldridge, 2010). To overcome this challenge, we leverage the temporal variation in remittance inflows by employing a continuous Difference-in-Differences (DiD) event-study design (Callaway et al., 2024), where our treatment is the pre-shock exposure to Florida, measured by $w_{m,FL}$.

4.2. Event-Study Specification

We use the following Poisson Pseudo Maximum Likelihood (PPML) event-study specification to analyze the impact of the Hurricane Ian shock on remittance inflows to Mexican municipalities:

$$\mathbb{E}[R_{mt} \mid \mathbf{X}_{mt}] = \exp \left(\alpha_m + \gamma_{s(m),t} + \sum_{k \neq -1} \beta_k \cdot (w_{m,FL} \cdot \mathbb{1}\{t - T_0 = k\}) \right) \quad (4.1)$$

The event-study specification allows us to estimate the dynamic treatment effects of the shock over time. We include municipality fixed effects (α_m) to control for all time-invariant, unobserved municipal characteristics that might otherwise introduce omitted variable bias. Additionally, we incorporate state-by-time fixed effects ($\gamma_{s(m),t}$), where

$s(m)$ denotes the specific Mexican state to which municipality m belongs. These fixed effects account for both macroeconomic shocks common to all regions and time-varying unobservables at the state level. The interaction term $w_{m,FL} \cdot \mathbb{1}\{t - T_0 = k\}$ captures the differential effect of the shock on Mexican municipalities based on their varying degrees of baseline exposure to Florida, where $w_{m,FL}$ is a measure of pre-shock exposure to Florida and $\mathbb{1}\{t - T_0 = k\}$ is an indicator for each time period relative to the shock. Therefore, the β_k coefficients represent the change in remittance inflows for a one-percent increase in Florida exposure at each time period k relative to the shock. Errors are clustered at the municipality level to account for potential correlation within municipalities over time.

4.3. Robustness Checks

Following Silva and Tenreyro, 2006, we employ the Poisson Pseudo Maximum Likelihood (PPML) estimator to address the heteroskedasticity and excess zeros, which are common features of remittance data at the municipality level. Unlike log-linear models, PPML yields consistent estimates regardless of heteroskedasticity, relying only on the correct specification of the conditional mean. However, heteroskedasticity affects inference, which we handle by clustering standard errors by municipality. This narrows our robustness checks to just two fundamental issues: the specification of the exponential conditional mean, and the data's mean-variance structure.

We address the first with a Ramsey RESET test as proposed by Ramsey (1969) and applied to this setting by Silva and Tenreyro (2006), which augments each specification with the square and cube of its fitted values and tests their joint nullity. A rejection signals that the assumed conditional mean is misspecified. We find that a log-linear OLS specification strongly rejects correct specification at all conventional levels ($W = 835.10, p < 0.001$), whereas the PPML specification does not reject the null ($W = 1.18, p = 0.308$).

The second question is answered by Figure 4.1, which plots, for each municipality, the variance of its pre-shock quarterly remittance inflows against their mean, both in logs. On these axes the Poisson benchmark is the line of slope one, along which the variance rises one-to-one with the mean. The cloud lies well above this line and follows a slope of approximately 1.546, so the variance scales faster than the mean, signaling overdispersion.

sion. This leaves PPML unaffected, since its consistency does not depend on the variance function. A log-linear model performs worse: the expected value of log inflows is not a linear function of the regressors, so estimating the model in logs by OLS does not recover the parameters of the conditional mean. The two diagnostics thereby support PPML as the baseline estimator.

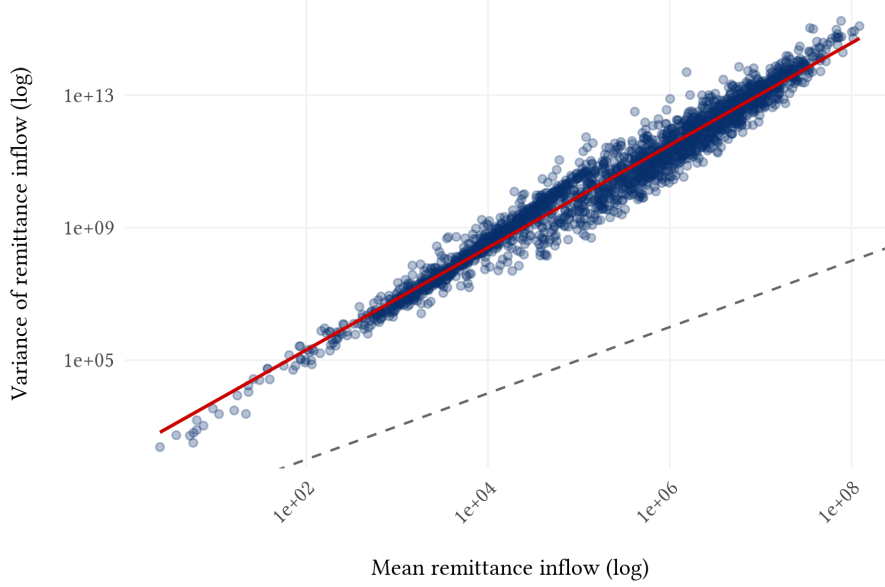


Figure 4.1: Mean-variance relationship of municipality remittance inflows. Each point represents a Mexican municipality; axes show the variance against the mean of its quarterly inflows over the pre-shock window. The solid line is the fitted relationship; the dashed line marks the Poisson equidispersion benchmark, where variance equals the mean.

4.4. Results

Figure 4.2 shows the result of the PPML event-study estimation, where the vertical dashed line represents the timing of the shock (Q4 2022). The pre-shock coefficients (β_k for $k < 0$) are close to zero and statistically insignificant, visually consistent with the parallel trends assumption underlying our DiD design. The point estimate in the quarter of the shock (Q4 2022) is negative and statistically significant at the 1% level, indicating that municipalities with higher exposure to Florida experienced a larger drop in remittance inflows immediately following the shock. Specifically, a one percentage point increase in Florida exposure is associated with a 0.16% decrease in remittance inflows in the quarter of the shock. In the subsequent quarters, the coefficients become slightly positive but statistically insignificant at the 5% level, suggesting that the initial

negative impact of the shock on remittance inflows dissipated over time.¹

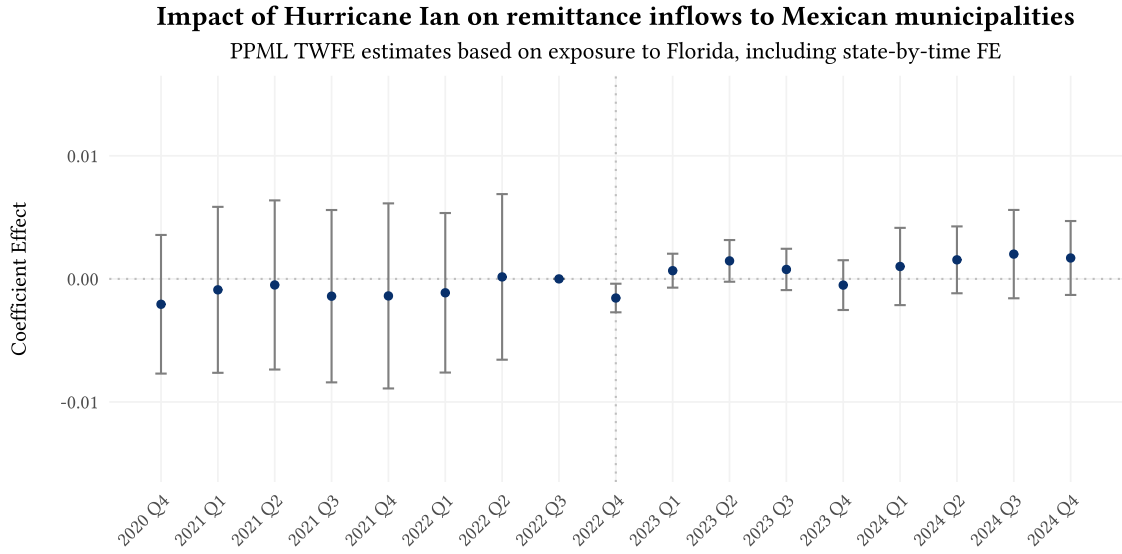


Figure 4.2: PPML Event Study Estimates

However, the absolute magnitude of this point estimate in Q4 2022 is comparable to some of the pre-trend fluctuations. Notably, there is a cyclical component that is not fully captured by the fixed effects, meaning that this drop could partly reflect a cyclical fluctuation rather than the pure effect of the shock. Nevertheless, the confidence intervals shrink noticeably during the shock period, suggesting that the hurricane injected a clear signal into the data. Moreover, this drop in our PPML TWFE estimates precedes the contraction observed in aggregate Florida remittance outflows, which do not visibly decline until Q1 and Q2 of 2023. Given the complexity of migration networks—and the possibility that workers in other US states altered their behavior to compensate for the Florida shock—these initial municipal-level results remain somewhat ambiguous. To disentangle these dynamics, Chapter ?? explores the presence of spillover effects from other states, which may be masking the true impact of the shock on remittance inflows to Mexican municipalities.

¹We assess robustness to violations of parallel trends using the relative-magnitudes approach of Rambachan and Roth (2023), reporting the breakdown value $\bar{M}$ (the largest pre-to-post violation ratio for which the effect stays significant; $\bar{M} = 1$ is a conventional lower bound). The baseline Florida-only effect breaks down at $\bar{M} = 0.15$. The sign and timing of the response are thus credible, but its magnitude is fragile, as expected given the dispersed, complex network of remittance flows, which motivates the spillover analysis that follows.

As a robustness check, we also estimate the same event-study specification using spatially clustered errors, following the methodology of Conley (1999), to account for potential spatial correlation in remittance inflow shocks affecting neighboring municipalities. The results, shown in Figure 4.3, are qualitatively similar to the baseline specification, with the main difference being the tighter confidence intervals in the preperiod as buffer distance increases, and slightly wider in the post-shock period. Given that the results are robust to both clustering methods, and that the spatial clustering specification doesn't produce positive definite variance-covariance matrices for buffer distances above 50km, we consider the municipality-level clustering as our baseline specification.

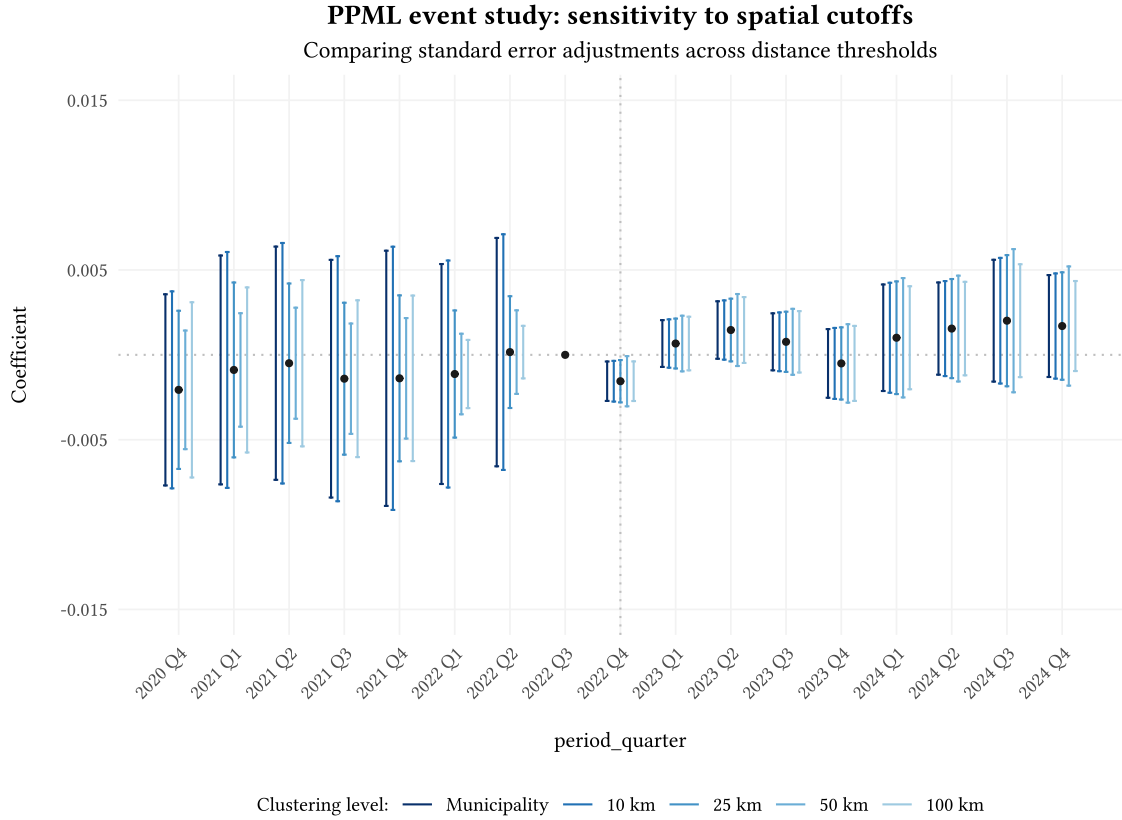


Figure 4.3: PPML Event Study Estimates with Conley Standard Errors for Different Buffer Distances

5. Investigating Spillover Effects

Therefore, we argue that our estimates are subject to general equilibrium effects. We base our analysis on the exposure-mapping logic of Aronow and Samii (2017) to define indirect exposure through alternative migrant networks. Rather than assuming that a

municipality's remittances respond only to Florida exposure, we allow the response to vary with exposure to Texas and California. While we do find economically plausible spillover effects from Texas, these findings are modest, as the shock is not large enough to disturb the whole of Florida's economy. Again, we emphasize on the idea of interpreting cautiously the impact of small, localized shocks.

5.1. Motivation

A fundamental requirement for standard causal inference models is the Stable Unit Treatment Value Assumption (SUTVA), formalized by Rubin (1980), which assumes that the potential outcomes of any unit depend exclusively on its own treatment assignment, excluding the possibility of any indirect spillover between units. In our context, U.S. states not directly exposed to a local shock might still be affected if control units are exposed to the treatment through indirect channels, leading to biased estimates (Aronow & Samii, 2017; Clarke, 2017). For instance, given that California and Texas all host a big proportion of Mexican migrants (each representing an average of 30% and 20% from the total, respectively), we hypothesize that these two states might have played a role in offsetting the negative Florida shock. If a Mexican municipality highly dependent on Florida remittances experiences an income shock due to Hurricane Ian, connected migrants residing in Texas might respond by increasing their transfers to compensate for the drop. If this was the case, the estimated Florida-exposure effect on total remittances would capture the net effect of the shock after within-network adjustment, rather than the direct fall in Florida-origin remittances.

5.2. Empirical Strategy

We test a possible compensation for the Florida shock from the states of Texas (TX) and California (CA) using an interaction strategy, expanding the PPML specification introduced in Chapter ??:

$$\mathbb{E}[R_{mt} \mid \mathbf{X}_{mt}] = \exp \left(\alpha_m + \gamma_{s(m),t} + \sum_{k \neq -1} \mathbb{1}\{t - T_0 = k\} \cdot \left[\beta_{1,k} w_{m,FL} + \beta_{2,k} w_{m,TX} + \beta_{3,k} (w_{m,FL} \times w_{m,TX}) \right] \right) \quad (5.1)$$

Consistent with the specification of Chapter ??, the equation above accounts for municipality fixed effects (α_m), Mexican state-quarter fixed effects ($\gamma_{s(m),t}$) and clustered standard errors at the municipality level. Note that an analogous specification can be estimated for the CA state, by replacing $w_{m, TX}$ with $w_{m, CA}$. Note that an analogous specification can be estimated for California by replacing $w_{m, TX}$ with $w_{m, CA}$.

To understand the mechanics of this spillover model, we can isolate the post-shock marginal effect of Florida exposure on expected municipal remittance inflows. Taking the partial derivative of the log conditional expectation with respect to Florida exposure yields:

$$\left. \frac{\partial \ln \mathbb{E}[R_{mt} \mid \mathbf{X}_{mt}]}{\partial w_{m, FL}} \right|_{\text{Post}=1} = \beta_1 + \beta_3 w_{m, TX} \quad (5.2)$$

Mathematically, β_1 represents the percentage change in remittances resulting from a one-unit increase in Florida exposure, evaluated strictly where Texas exposure is zero ($w_{m, TX} = 0$). On the other hand, the coefficient of the interaction term, β_3 measures how the post-shock effect of Florida exposure changes for every one-unit increase in Texas exposure. Intuitively, if $\beta_3 > 0$, the Texas network compensates for the Florida's drop in remittances, mitigating the negative impact of the shock. Conversely, if $\beta_3 < 0$, the Texas network amplifies the negative effect of the Florida shock, potentially due to out-migration costs.

Because β_1 measures how the hurricane impacted municipalities that rely on Florida but possess absolutely no alternative migration network in Texas, which might not represent a meaningful real-world baseline for many municipalities, we recenter the $w_{m, TX}$ variable at different percentiles of exposure to Texas. For example, if we recenter Texas exposure at its 90th percentile, the newly estimated β_1 gives us the isolated impact of the Florida shock on highly networked municipalities. By comparing the β_1 estimates across models recentered at low versus high percentiles of Texas exposure, we can formally test how aggressively the presence of a secondary network alters the impact of the Florida shock on remittance inflows.

5.3. Results

Figure 5.1 shows the dynamic evolution of the β_3 coefficient. For both CA and TX, the pre-trends are flat before the Hurricane event, suggesting parallel trends that would validate our identification strategy. As for the coefficients post-shock, we notice different patterns across the two states. In the CA case, we observe that the panel stays essentially flat around zero, which suggests that no significant spillover effects were found. As for TX, we see a slightly negative, significant and persistent decline of the coefficient, meaning that the more exposed a given municipality is to TX, the more pronounced is the negative post-shock effect of the FL exposure on remittance patterns. In other words, higher TX exposure is associated to a larger net decline in remittances after the FL shock.

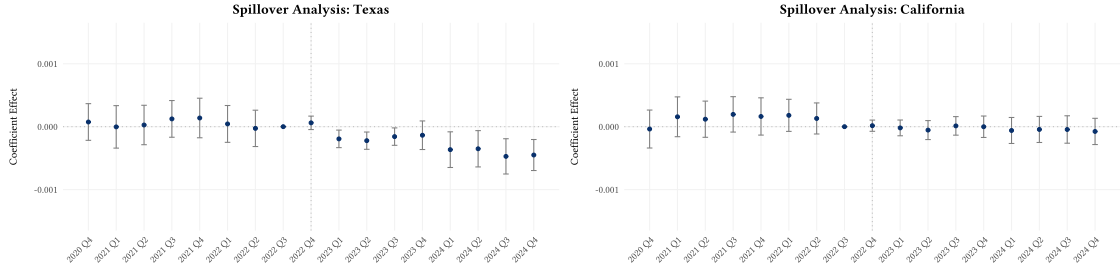


Figure 5.1: Evolution of the coefficient β_3 over time for the interaction between FL and TX (left) and FL and CA (right). The vertical dashed line indicates the timing of the Hurricane Ian shock. The shaded area represents the 95% confidence interval.

Since this result a priori does not support the compensatory spillover hypothesis of the model, we next aim to explore how the post-shock FL effect on remittances varies according to different levels of TX exposure. Figure 5.2 plots the $\tilde{\beta}_1$ coefficient, constructed by recentering the values of $Exposure_m^{TX}$ on the 10th and 90th percentiles of the distribution of TX exposure across municipalities. The net effect of the FL shock on remittances is more negative for municipalities with higher levels of TX exposure, which is consistent with the negative and statistically significant β_3 coefficient found above. This reinforces the idea that instead of a compensatory effect, we observe an amplification of the negative impact of the FL shock on remittances is more pronounced in municipalities that are also highly exposed to TX.

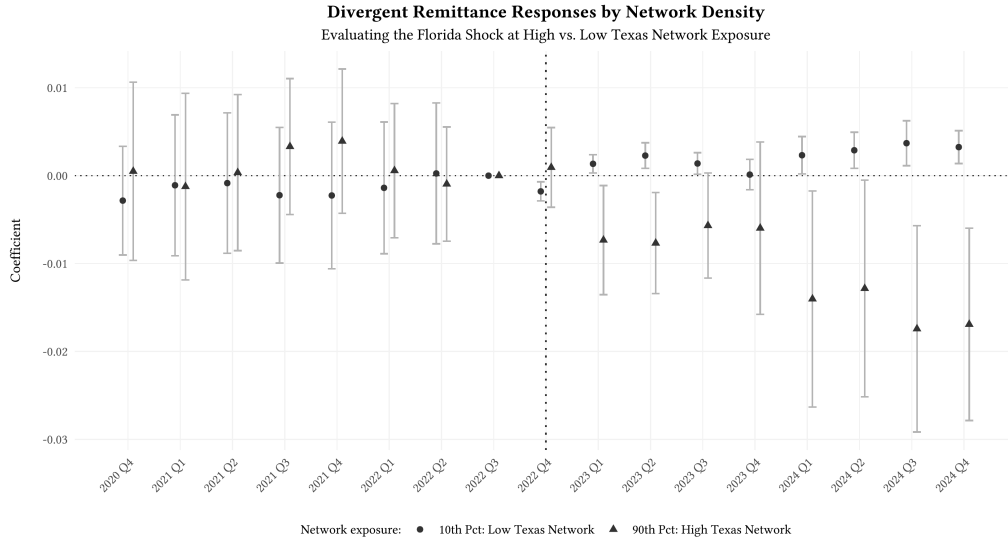


Figure 5.2: Net Florida shock evaluated at low and high levels of Texas exposure. The figure compares the implied post-shock Florida effect for municipalities at the 10th and 90th percentiles of Texas exposure.

One possible interpretation at the 90th percentile of Texas exposure is internal mobility within the US. For municipalities with strong pre-existing connections to Texas, these established networks may allow migrants to easily relocate from Florida to Texas after the hurricane. This behavior aligns with Fussell et al. (2023), who document immediate and persistent out-migration from disaster zones following Hurricanes Katrina and Rita. In contrast, migrants from municipalities at the 10th percentile of Texas exposure may find it much harder to relocate, thus remaining in Florida. Paradoxically, Figure 5.2 suggests, that those who stay in Florida after the hurricane can participate to the reconstruction boom that follows, experiencing a wage increase and therefore an increase in remittances from the lenses of our model discussed in Chapter 2. Conversely, those who relocate to Texas might face a decrease in their income due to frictions in the employment process or relocation costs.

This dynamic explains the timing observed in our event-study estimates. For municipalities highly connected to Texas, remittance inflows do not drop significantly in the exact quarter of the shock. Instead, the relative decline emerges and deepens in the subsequent quarters, particularly when compared to low-exposure municipalities. This delayed drop is consistent with the frictions of the migration process, while the delayed increase for the low-exposure municipalities reflects the participation to the reconstruction boom.

6. Conclusion

6.1. Summary of Findings

The main objective of this study was to investigate the impact of a local shock, specifically Hurricane Ian, on remittance patterns from U.S. states to Mexican municipalities. Our analysis revealed a significant negative effect of Florida exposure on remittances, indicating that municipalities with higher exposure to Florida experienced a more pronounced decline in remittance inflows following the hurricane. In fact, the estimated effect suggests that a 1% increase in Florida exposure from the Mexican municipality is associated with a 0.16% decrease in remittances, which is a substantial impact given the importance of remittances for many Mexican households. This findings are consistent with the theoretical approach provided in the model, which predicts that a negative shock in a major U.S. destination can lead to a decline in remittances to the origin municipality, especially for those with higher exposure to the affected state.

To investigate the potential spillover effects from other major U.S. destinations, coherently with what the model would suggest, we explored the role of Texas and California as alternative migrant networks that could potentially compensate for the negative shock in Florida. However, our analysis did not find evidence of a compensatory effect from California, while for Texas we observed an amplification effect, where the negative impact of the Florida shock on remittances was more pronounced for municipalities with higher exposure to Texas. This suggests that instead of mitigating the negative shock from Florida, Texas exposure may have exacerbated it. More concretely, the estimated spillover effect from Texas indicates that a significant 1 percentage-point increase in Texas exposure amplifies the negative post-shock effect of Florida exposure by approximately 0.04%.

To theoretically explain this results the model should account for not only the Mexican families but also the migrant networks in the U.S. hitted by a shock, to be potential transfer recievers. The altruition logic in the model would also apply in the shock setting, treating those Mexican U.S households as needy remittances recievers. The amplification effect observed would be consistent under the assumption that **TO BE COMPLETED**

Finally, our findings suggest that Caballero et al., 2023 results on the persistence of remittance patterns after a shock might not be generalizable to all contexts, specially when we analyse a localized event. Their measured impacts may combine destination-country income shocks with origin-country demand effects generated by the global nature of the Great Recession, meanwhile, the shock used by this paper is clean and localized, allowing us to isolate the direct effect of the shock on remittance patterns without confounding factors.

6.2. Policy Implications

The findings of this study have important policy implications for both Mexican and U.S. policymakers. For Mexican policymakers, the significant decline in remittances following the Florida shock highlights the vulnerability of Mexican households that rely heavily on remittance inflows although a relevant finding is also the non persistence of the Florida exposed remittances decline. This suggests a need for policies that can provide immediate alternative sources of income or financial support to affected households, especially in municipalities with high exposure to U.S. destinations but without the necessity to give a long term support. For U.S. policymakers, the results underscore the importance of considering the broader economic impacts of local shocks, such as natural disasters, on migrant communities and their families back home. Policies aimed at providing support to affected migrants could help mitigate the negative spillover effects on remittance flows by reducing the internal transfers between connected migrants across states. **NOTE THAT WE DONT TALK ABOUT INTERNAL TRANSFERS ANYMORE! ID TAKE AWAY THE WHOLE SENTENCE**

6.3. Limitations and Future Research

While this study provides valuable insights into the impact of local shocks on remittance patterns, it also has several limitations that should be acknowledged. First, our analysis is based on a single shock event (Hurricane Ian), which may limit the generalizability of our findings to other types of shocks or different contexts. Future research could explore the effects of various shocks, such as economic downturns or policy changes, on remittance patterns to provide a more comprehensive understanding of the dynamics at

play. Second, our measure of exposure is based on the percentage of migrants from each municipality residing in specific U.S. states, which may not fully capture the complexity of migrant networks and their influence on remittance behavior. Future studies could incorporate more detailed data on migrant networks, including social ties and communication patterns, to better understand how these factors mediate the impact of shocks on remittances. Finally, while we explored potential spillover effects from Texas and California, there may be other states or factors that contribute to the observed patterns. Future research could investigate additional channels through which shocks in one state may affect remittance flows from other states, such as labor market conditions or policy responses in those states.

I would also include: the fact that remittance sent vary depending on how many immigrants go in and out of the states, which is data we do not have (and that could refine our analysis).

References

- Albert, C., & Monras, J. (2022). Immigration and spatial equilibrium: The role of expenditures in the country of origin. *American Economic Review*, 112(11), 3763–3802. <https://doi.org/10.1257/aer.20211241>
- Alcaraz, C., Chiquiar, D., & Salcedo, A. (2012). Remittances, schooling, and child labor in Mexico. *Journal of Development Economics*, 97(1), 156–165. <https://doi.org/10.1016/j.jdeveco.2010.11.004>
- Amuedo-Dorantes, C., & Pozo, S. (2006). Remittances as insurance: Evidence from mexican immigrants. *Journal of Population Economics*, 19, 227–254. <https://doi.org/10.1007/s00148-006-0079-6>
- Aronow, P. M., & Samii, C. (2017). Estimating average causal effects under general interference, with application to a social network experiment. *The Annals of Applied Statistics*, 11(4), 1912–1947. <https://doi.org/10.1214/16-AOAS1005>
- Banco de México. (2026a). *Estado de origen de los ingresos por remesas provenientes de Estados Unidos (CE168)* [Sistema de Información Económica (SIE), Banco de México]. Retrieved May 26, 2026, from <http://www.banxico.org.mx/SieInternet/consultarDirectorioInternetAction.do?accion=consultarCuadro&idCuadro=CE168&locale=es>
- Banco de México. (2026b). *Ingresos por remesas familiares (CA11)* [Sistema de Información Económica (SIE), Banco de México]. Retrieved May 26, 2026, from <https://www.banxico.org.mx/SieInternet/consultarDirectorioInternetAction.do?accion=consultarCuadroAnalitico&idCuadro=CA11§or=1&locale=es>
- Banco de México. (2026c). *Ingresos por remesas, distribución por municipio (CE166)* [Sistema de Información Económica (SIE), Banco de México]. Retrieved May 26, 2026, from <https://www.banxico.org.mx/SieInternet/consultarDirectorioInternetAction.do?accion=consultarCuadro&idCuadro=CE166&locale=es>
- Bettin, G., Jallow, A., & Zazzaro, A. (2025). Responding to natural disasters: What do monthly remittance data tell us? *Journal of Development Economics*, 174, 103413. <https://doi.org/10.1016/j.jdeveco.2024.103413>
- Boustan, L. P., Kahn, M. E., Rhode, P. W., & Yanguas, M. L. (2020). The effect of natural disasters on economic activity in US counties: A century of data. *Journal of Urban Economics*, 118, 103257. <https://doi.org/10.1016/j.jue.2020.103257>
- Bucci, L., Alaka, L., Hagen, A., Delgado, S., & Beven, J. (2026, March 12). *Tropical cyclone report: Hurricane Ian (AL092022), 23–30 September 2022* (tech. rep.). National Hurricane Center, National Oceanic and Atmospheric Administration. Retrieved May 26, 2026, from https://www.nhc.noaa.gov/data/tcr/AL092022_Ian.pdf
- Butts, K. (2023). Difference-in-differences estimation with spatial spillovers. <https://arxiv.org/abs/2105.03737>
- Caballero, M. E., Cadena, B. C., & Kovak, B. K. (2018). Measuring geographic migration patterns using Matriculas Consulares. *Demography*, 55(3), 1119–1145. <https://doi.org/10.1007/s13524-018-0675-6>
- Caballero, M. E., Cadena, B. C., & Kovak, B. K. (2023). The international transmission of local economic shocks through migrant networks. *Journal of International Economics*, 145, 103832. <https://doi.org/10.1016/j.jinteco.2023.103832>
- Callaway, B., Goodman-Bacon, A., & Sant’Anna, P. H. (2024). Difference-in-differences with a continuous treatment. *Journal of Econometrics*, 241(1), 105746.
- Card, D. (2001). Immigrant inflows, native outflows, and the local labor market impacts of higher immigration. *Journal of Labor Economics*, 19(1), 22–64. <https://doi.org/10.1086/209979>

- Clarke, D. (2017). *Estimating difference-in-differences in the presence of spillovers* (MPRA Paper No. 81604). University Library of Munich. <https://mpra.ub.uni-muenchen.de/81604/>
- Conley, T. G. (1999). Gmm estimation with cross sectional dependence. *Journal of Econometrics*, 92(1), 1–45.
- Fussell, E., DeWaard, J., & Curtis, K. J. (2023). Environmental migration as short- or long-term differences from a trend: A case study of hurricanes katrina and rita effects on out-migration in the gulf of mexico. *Int. Migr.*, 61(5), 60–74.
- Gordillo de Anda, G., & Plassot, T. (2017). Movilidad interna en México: Tendencias recientes y patrones regionales. *Economía, Sociedad y Territorio*, 17(55), 1021–1050.
- Handlin, L., Krontoft, M., & Testa, W. (2002). Remittances and the unbanked (special issue). *Chicago Fed Letter*, (175a).
- Hudgens, M. G., & Halloran, M. E. (2008). Toward causal inference with interference. *Journal of the American Statistical Association*, 103(482), 832–842. <https://doi.org/10.1198/016214508000000292>
- Instituto de los Mexicanos en el Exterior. (2026). *Estadísticas: Información estadística de la población mexicana en el exterior*. Secretaría de Relaciones Exteriores. Retrieved April 1, 2026, from <https://ime.gob.mx/estadisticas>
- Melitz, M. J. (2003). The impact of trade on intra-industry reallocations and aggregate industry productivity. *Econometrica*, 71(6), 1695–1725. <https://doi.org/10.1111/1468-0262.00467>
- Mora-Rivera, J., & van Gameren, E. (2021). The impact of remittances on food insecurity: Evidence from Mexico. *World Development*, 140, 105349. <https://doi.org/10.1016/j.worlddev.2020.105349>
- Munshi, K. (2003). Networks in the modern economy: Mexican migrants in the U.S. labor market. *The Quarterly Journal of Economics*, 118(2), 549–599. <https://doi.org/10.1162/003355303321675455>
- Partida Bush, V. (2014). Migración interna en México y urbanización: Perspectivas recientes. *Migración y Desarrollo*, 8(2), 11–34.
- Rambachan, A., & Roth, J. (2023). A more credible approach to parallel trends. *The Review of Economic Studies*, 90(5), 2555–2591. <https://doi.org/10.1093/restud/rdad018>
- Ramsey, J. B. (1969). Tests for specification errors in classical linear least squares regression analysis. *Journal of the Royal Statistical Society B*, 31(2), 350–371.
- Rubin, D. B. (1980). Randomization analysis of experimental data: The Fisher randomization test comment. *Journal of the American Statistical Association*, 75(371), 591–593. <https://doi.org/10.1080/01621459.1980.10477517>
- Silva, J. M. C. S., & Tenreyro, S. (2006). The log of gravity. *The Review of Economics and Statistics*, 88(4), 641–658. <https://doi.org/10.1162/rest.88.4.641>
- Taylor, J. E., Mora, J., Adams, R. H., & Lopez-Feldman, A. (2005). *Remittances, inequality and poverty: Evidence from rural Mexico* (Working Paper No. 05-003). University of California, Davis, Department of Agricultural and Resource Economics. <https://doi.org/10.22004/ag.econ.60287>
- Wooldridge, J. M. (2010). *Econometric analysis of cross section and panel data* (2nd). MIT Press.
- World Bank. (2024). *Personal remittances, received (current US\$)* [World Development Indicators, series BX.TRF.PWKR.CD.DT]. Retrieved May 26, 2026, from <https://data.worldbank.org/indicator/BX.TRF.PWKR.CD.DT?locations=MX>
- Yang, D., & Choi, H. (2007). Are remittances insurance? evidence from rainfall shocks in the Philippines. *The World Bank Economic Review*, 21(2), 219–248. <https://doi.org/10.1093/wber/lhm003>